

FDTD Equations Reference

From `fdtd_brain_simulation_engine.py`

This document lists and labels all physical and mathematical equations implemented in the 3D FDTD brain simulation engine. Equation numbers match the order of use in the code (time step $\rightarrow$ materials $\rightarrow$ field updates $\rightarrow$ SAR $\rightarrow$ thermal).

1 Derivation: Continuous to Discrete (Methodology)

Notation: time index n , half-integer time $n + \frac{1}{2}$; spatial indices (i, j, k) on a uniform grid with $\Delta x = \Delta y = \Delta z$; E and H staggered on the Yee grid. The flow is: continuous PDE $\rightarrow$ approximation rules $\rightarrow$ substitute $\rightarrow$ rearrange $\rightarrow$ explicit update. Aligned with the methodology in the thesis (official_paper.tex, FDTD Solver Implementation).

1.1 Time step

Stability for the 3D explicit scheme requires the Courant condition. Rule: maximum time step set by the minimum wave-propagation distance across one cell.

$$\Delta t = \frac{F}{c_0 \sqrt{1/(\Delta x)^2 + 1/(\Delta y)^2 + 1/(\Delta z)^2}}, \quad c_0 = \frac{1}{\sqrt{\mu_0 \epsilon_0}}. \quad (1)$$

F is the Courant factor (e.g. 0.99).

1.2 Electric field: from Ampere's law to explicit update

Implementation order (per time step): update $\mathbf{D}$ from $\mathbf{H}$, then $\mathbf{E}$ and $\mathbf{I}$ from $\mathbf{D}$, then $\mathbf{H}$ from $\mathbf{E}$. Thus the E-update uses $\mathbf{H}$ at the previous half-step ($\mathbf{H}^{n-1/2}$), and the H-update uses the newly computed $\mathbf{E}^{n+1}$.

1. Continuous equation (from Maxwell–Ampère). Start from Maxwell–Ampère in differential form:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}. \quad (2)$$

In a linear, isotropic, lossy dielectric (no impressed current), use the constitutive relations $\mathbf{J} = \sigma \mathbf{E}$ and $\mathbf{D} = \epsilon \mathbf{E}$ with $\epsilon = \epsilon_r \epsilon_0$, giving

$$\nabla \times \mathbf{H} = \sigma \mathbf{E} + \epsilon \frac{\partial \mathbf{E}}{\partial t}. \quad (3)$$

Write the curl explicitly in Cartesian components:

$$\nabla \times \mathbf{H} = \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) \hat{\mathbf{x}} + \left(\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} \right) \hat{\mathbf{y}} + \left(\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} \right) \hat{\mathbf{z}}. \quad (4)$$

Taking the x -component gives $(\nabla \times \mathbf{H})_x = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}$, so

$$\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} = \sigma E_x + \varepsilon \frac{\partial E_x}{\partial t}. \quad (5)$$

Solve for $\frac{\partial E_x}{\partial t}$ to obtain the form used for discretization:

$$\frac{\partial E_x}{\partial t} = \frac{1}{\varepsilon} \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} \right) - \frac{\sigma}{\varepsilon} E_x. \quad (6)$$

Rule: Maxwell–Ampère + $\mathbf{J} = \sigma \mathbf{E} + \mathbf{D} = \varepsilon \mathbf{E} + x\text{-component of } \nabla \times \mathbf{H}$.

2. Approximation rules. Central difference in time (leapfrog); central difference in space (Yee); conductivity term approximated by the average of E_x at two half-steps:

$$\begin{aligned} \frac{\partial F}{\partial t} &\approx \frac{F^{n+1/2} - F^{n-1/2}}{\Delta t} \\ \frac{\partial G}{\partial y} &\approx \frac{G(j + \frac{1}{2}) - G(j - \frac{1}{2})}{\Delta y} \\ \frac{\partial G}{\partial z} &\approx \frac{G(k + \frac{1}{2}) - G(k - \frac{1}{2})}{\Delta z} \\ E_x &\approx \frac{E_x^{n+1/2} + E_x^{n-1/2}}{2} \end{aligned} \quad (7)$$

3. Substitute into Continuous Equation Eq. (6).

$$\begin{aligned} \frac{E_x^{n+1/2}(i, j, k) - E_x^{n-1/2}(i, j, k)}{\Delta t} &= \frac{1}{\varepsilon} \left(\frac{H_z^{n-1/2}(i, j, k) - H_z^{n-1/2}(i, j - 1, k)}{\Delta y} \right. \\ &\quad \left. - \frac{H_y^{n-1/2}(i, j, k) - H_y^{n-1/2}(i, j, k - 1)}{\Delta z} \right) \\ &\quad - \frac{\sigma}{\varepsilon} \frac{E_x^{n+1/2}(i, j, k) + E_x^{n-1/2}(i, j, k)}{2}. \end{aligned} \quad (8)$$

Rule: finite-difference substitution using (7); curl $\mathbf{H}$ evaluated at $n - \frac{1}{2}$ to match implementation order (E updated from D, D from $\mathbf{H}^{n-1/2}$).

4. Multiply by Δt and rearrange. Multiply both sides of (8) by Δt . The left-hand side becomes $E_x^{n+1/2} - E_x^{n-1/2}$. On the right, the conductivity term expands to $-\frac{\sigma \Delta t}{2\varepsilon} E_x^{n+1/2} - \frac{\sigma \Delta t}{2\varepsilon} E_x^{n-1/2}$. Collect all terms containing $E_x^{n+1/2}$ on the left and all terms containing $E_x^{n-1/2}$ or the H -field curls on the right; then factor out $E_x^{n+1/2}$ and $E_x^{n-1/2}$ to obtain

$$\begin{aligned} E_x^{n+1/2} \left(1 + \frac{\sigma \Delta t}{2\varepsilon} \right) &= E_x^{n-1/2} \left(1 - \frac{\sigma \Delta t}{2\varepsilon} \right) \\ &\quad + \frac{\Delta t}{\varepsilon} \left(\frac{H_z^{n-1/2}(i, j, k) - H_z^{n-1/2}(i, j - 1, k)}{\Delta y} \right. \\ &\quad \left. - \frac{H_y^{n-1/2}(i, j, k) - H_y^{n-1/2}(i, j, k - 1)}{\Delta z} \right). \end{aligned} \quad (9)$$

Rule: algebraic rearrangement (multiply by Δt , expand, collect unknowns $E_x^{n+1/2}$ on the left, knowns $E_x^{n-1/2}$ and H on the right, then factor).

5. Explicit update for E_x . Divide by $(1 + \frac{\sigma\Delta t}{2\varepsilon})$:

$$\begin{aligned} E_x^{n+1/2}(i, j, k) = & \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)} E_x^{n-1/2}(i, j, k) \\ & + \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)} \left(\frac{H_z^{n-1/2}(i, j, k) - H_z^{n-1/2}(i, j-1, k)}{\Delta y} \right. \\ & \left. - \frac{H_y^{n-1/2}(i, j, k) - H_y^{n-1/2}(i, j, k-1)}{\Delta z} \right). \end{aligned} \quad (10)$$

Rule: solve for $E_x^{n+1/2}$; uses $\mathbf{H}^{n-1/2}$ (implementation order: D from H, then E from D, then H from E). The same methodology applied to the y - and z -components yields the electric field updates given in the thesis (FDTD Solver Implementation) with spatial derivatives replaced by the corresponding central differences on the Yee grid.

Index reformulation (Yee staggering). The explicit updates are derived at a generic index (i, j, k) with half-integer time for E ($n \pm \frac{1}{2}$) and H at $n - \frac{1}{2}$. In the thesis and code, the same quantities are stored with explicit Yee staggering: E at integer time $n, n+1$ and at half-offsets in the direction of the component; H at half-integer time and at half-offsets in the two directions perpendicular to the component. The following mappings translate generic indices to Yee indices for each field component.

Electric field (curl $\mathbf{H} \rightarrow \mathbf{E}$). E_x -update (uses H_z, H_y):

$$\begin{aligned} E_x^{n+1/2}(i, j, k) & \longleftrightarrow E_x^{n+1}(i + \frac{1}{2}, j, k), & E_x^{n-1/2}(i, j, k) & \longleftrightarrow E_x^n(i + \frac{1}{2}, j, k), \\ H_z^{n-1/2}(i, j, k) & \longleftrightarrow H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k), & H_z^{n-1/2}(i, j-1, k) & \longleftrightarrow H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j - \frac{1}{2}, k), \\ H_y^{n-1/2}(i, j, k) & \longleftrightarrow H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}), & H_y^{n-1/2}(i, j, k-1) & \longleftrightarrow H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k - \frac{1}{2}). \end{aligned} \quad (11)$$

E_y -update (uses H_x, H_z):

$$\begin{aligned} E_y^{n+1/2}(i, j, k) & \longleftrightarrow E_y^{n+1}(i, j + \frac{1}{2}, k), & E_y^{n-1/2}(i, j, k) & \longleftrightarrow E_y^n(i, j + \frac{1}{2}, k), \\ H_x^{n-1/2}(i, j, k) & \longleftrightarrow H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}), & H_x^{n-1/2}(i, j, k-1) & \longleftrightarrow H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k - \frac{1}{2}), \\ H_z^{n-1/2}(i, j, k) & \longleftrightarrow H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k), & H_z^{n-1/2}(i-1, j, k) & \longleftrightarrow H_z^{n-\frac{1}{2}}(i - \frac{1}{2}, j + \frac{1}{2}, k). \end{aligned} \quad (12)$$

E_z -update (uses H_y, H_x):

$$\begin{aligned} E_z^{n+1/2}(i, j, k) & \longleftrightarrow E_z^{n+1}(i, j, k + \frac{1}{2}), & E_z^{n-1/2}(i, j, k) & \longleftrightarrow E_z^n(i, j, k + \frac{1}{2}), \\ H_y^{n-1/2}(i, j, k) & \longleftrightarrow H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}), & H_y^{n-1/2}(i-1, j, k) & \longleftrightarrow H_y^{n-\frac{1}{2}}(i - \frac{1}{2}, j, k + \frac{1}{2}), \\ H_x^{n-1/2}(i, j, k) & \longleftrightarrow H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}), & H_x^{n-1/2}(i, j-1, k) & \longleftrightarrow H_x^{n-\frac{1}{2}}(i, j - \frac{1}{2}, k + \frac{1}{2}). \end{aligned} \quad (13)$$

Magnetic field (curl $\mathbf{E} \rightarrow \mathbf{H}$). H_x -update (uses E_z, E_y); time remains half-integer $n \pm \frac{1}{2}$:

$$\begin{aligned} H_x^{n+1/2}(i, j, k) &\longleftrightarrow H_x^{n+\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}), & H_x^{n-1/2}(i, j, k) &\longleftrightarrow H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}), \\ E_z^{n+1}(i, j + 1, k) &\longleftrightarrow E_z^{n+1}(i, j + 1, k + \frac{1}{2}), & E_z^{n+1}(i, j, k) &\longleftrightarrow E_z^{n+1}(i, j, k + \frac{1}{2}), \\ E_y^{n+1}(i, j, k + 1) &\longleftrightarrow E_y^{n+1}(i, j + \frac{1}{2}, k + 1), & E_y^{n+1}(i, j, k) &\longleftrightarrow E_y^{n+1}(i, j + \frac{1}{2}, k). \end{aligned} \quad (14)$$

H_y -update (uses E_x, E_z):

$$\begin{aligned} H_y^{n+1/2}(i, j, k) &\longleftrightarrow H_y^{n+\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}), & H_y^{n-1/2}(i, j, k) &\longleftrightarrow H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}), \\ E_x^{n+1}(i, j, k + 1) &\longleftrightarrow E_x^{n+1}(i + \frac{1}{2}, j, k + 1), & E_x^{n+1}(i, j, k) &\longleftrightarrow E_x^{n+1}(i + \frac{1}{2}, j, k), \\ E_z^{n+1}(i + 1, j, k) &\longleftrightarrow E_z^{n+1}(i + 1, j, k + \frac{1}{2}), & E_z^{n+1}(i, j, k) &\longleftrightarrow E_z^{n+1}(i, j, k + \frac{1}{2}). \end{aligned} \quad (15)$$

H_z -update (uses E_y, E_x):

$$\begin{aligned} H_z^{n+1/2}(i, j, k) &\longleftrightarrow H_z^{n+\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k), & H_z^{n-1/2}(i, j, k) &\longleftrightarrow H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k), \\ E_y^{n+1}(i + 1, j, k) &\longleftrightarrow E_y^{n+1}(i + 1, j + \frac{1}{2}, k), & E_y^{n+1}(i, j, k) &\longleftrightarrow E_y^{n+1}(i, j + \frac{1}{2}, k), \\ E_x^{n+1}(i, j + 1, k) &\longleftrightarrow E_x^{n+1}(i + \frac{1}{2}, j + 1, k), & E_x^{n+1}(i, j, k) &\longleftrightarrow E_x^{n+1}(i + \frac{1}{2}, j, k). \end{aligned} \quad (16)$$

Substituting the E_x mapping (11) into (10) (and the E_y, E_z mappings (12)–(13) into the analogous explicit updates) reproduces the Yee-staggered electric-field updates in the thesis. The H_x, H_y, H_z mappings (14)–(16) likewise reproduce the magnetic-field updates in fully staggered form. Explicitly, the E -updates in fully staggered index form are

$$\begin{aligned} E_x^{n+1}(i + \frac{1}{2}, j, k) &= \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)} E_x^n(i + \frac{1}{2}, j, k) \\ &+ \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)} \left(\frac{H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k) - H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j - \frac{1}{2}, k)}{\Delta y} \right. \\ &\quad \left. - \frac{H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}) - H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k - \frac{1}{2})}{\Delta z} \right), \end{aligned} \quad (17)$$

$$\begin{aligned} E_y^{n+1}(i, j + \frac{1}{2}, k) &= \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)} E_y^n(i, j + \frac{1}{2}, k) \\ &+ \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)} \left(\frac{H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}) - H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k - \frac{1}{2})}{\Delta z} \right. \\ &\quad \left. - \frac{H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k) - H_z^{n-\frac{1}{2}}(i - \frac{1}{2}, j + \frac{1}{2}, k)}{\Delta x} \right), \end{aligned} \quad (18)$$

$$\begin{aligned} E_z^{n+1}(i, j, k + \frac{1}{2}) &= \frac{1 - \sigma\Delta t/(2\varepsilon)}{1 + \sigma\Delta t/(2\varepsilon)} E_z^n(i, j, k + \frac{1}{2}) \\ &+ \frac{\Delta t/\varepsilon}{1 + \sigma\Delta t/(2\varepsilon)} \left(\frac{H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}) - H_y^{n-\frac{1}{2}}(i - \frac{1}{2}, j, k + \frac{1}{2})}{\Delta x} \right. \\ &\quad \left. - \frac{H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}) - H_x^{n-\frac{1}{2}}(i, j - \frac{1}{2}, k + \frac{1}{2})}{\Delta y} \right). \end{aligned} \quad (19)$$

These are algebraically equivalent to (10) (and its y, z analogues), but written with the explicit Yee staggering of indices used in the thesis.

1.3 Flux density and auxiliary summation (equivalent formulation)

Introduce the flux density and the integrated conduction current:

$$\mathbf{D} = \varepsilon \mathbf{E}, \quad \frac{\partial \mathbf{I}}{\partial t} = \sigma \mathbf{E}. \quad (20)$$

Then Maxwell–Ampère becomes

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{I}}{\partial t} + \frac{\partial \mathbf{D}}{\partial t}. \quad (21)$$

In sampled time, define the summation parameter

$$I^{n-1} = \frac{\sigma \Delta t}{\varepsilon_0} \sum_{i=0}^{n-1} E^i, \quad D^n = \varepsilon_r E^n + \frac{\sigma \Delta t}{\varepsilon_0} \sum_{i=0}^{n-1} E^i. \quad (22)$$

Rule: sampled-time constitutive relation used to avoid storing the full history $\{E^i\}$ by carrying I recursively.

$$E^n = \frac{D^n - I^{n-1}}{\varepsilon_r + \sigma \Delta t / \varepsilon_0}, \quad I^n = I^{n-1} + \frac{\sigma \Delta t}{\varepsilon_0} E^n, \quad (23)$$

Define the per-voxel coefficients (often stored per component in code):

$$g_a = \frac{1}{\varepsilon_r + \sigma \Delta t / \varepsilon_0}, \quad g_b = \frac{\sigma \Delta t}{\varepsilon_0}. \quad (24)$$

Then the updates emphasize the one-line affine form:

$$E = g_a(D - I), \quad I^n = I^{n-1} + g_b E^n. \quad (25)$$

Rule: equivalent to the explicit lossy E -update above; avoids storing past E by updating I recursively.

1.4 Displacement $\mathbf{D}$: $\partial \mathbf{D} / \partial t = \nabla \times \mathbf{H}$

Continuous (component form of $\partial \mathbf{D} / \partial t = \nabla \times \mathbf{H}$):

$$\frac{\partial D_x}{\partial t} = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z}, \quad \frac{\partial D_y}{\partial t} = \frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x}, \quad \frac{\partial D_z}{\partial t} = \frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y}. \quad (26)$$

Rule: forward difference in time; central difference in space on the Yee grid. Per-step order: $\mathbf{D}$ is updated first using $\mathbf{H}^{n-1/2}$ (previous half-step).

$$\begin{aligned} D_x^{n+1}(i, j, k) &= D_x^n(i, j, k) \\ &+ \Delta t \left(\frac{H_z^{n-\frac{1}{2}}(i, j, k) - H_z^{n-\frac{1}{2}}(i, j-1, k)}{\Delta y} \right. \\ &\quad \left. - \frac{H_y^{n-\frac{1}{2}}(i, j, k) - H_y^{n-\frac{1}{2}}(i, j, k-1)}{\Delta z} \right) \end{aligned} \quad (27)$$

$$\begin{aligned} D_y^{n+1}(i, j, k) &= D_y^n(i, j, k) \\ &+ \Delta t \left(\frac{H_x^{n-\frac{1}{2}}(i, j, k) - H_x^{n-\frac{1}{2}}(i, j, k-1)}{\Delta z} \right. \\ &\quad \left. - \frac{H_z^{n-\frac{1}{2}}(i, j, k) - H_z^{n-\frac{1}{2}}(i-1, j, k)}{\Delta x} \right) \end{aligned} \quad (28)$$

$$\begin{aligned} D_z^{n+1}(i, j, k) &= D_z^n(i, j, k) \\ &+ \Delta t \left(\frac{H_y^{n-\frac{1}{2}}(i, j, k) - H_y^{n-\frac{1}{2}}(i-1, j, k)}{\Delta x} \right. \\ &\quad \left. - \frac{H_x^{n-\frac{1}{2}}(i, j, k) - H_x^{n-\frac{1}{2}}(i, j-1, k)}{\Delta y} \right) \end{aligned} \quad (29)$$

Component-wise E and I updates (flux formulation), made explicit for each Cartesian component:

$$\begin{aligned} E_x^n(i, j, k) &= g_{a,x}(i, j, k) \left(D_x^n(i, j, k) - I_x^{n-1}(i, j, k) \right), \\ I_x^n(i, j, k) &= I_x^{n-1}(i, j, k) + g_{b,x}(i, j, k) E_x^n(i, j, k), \end{aligned} \quad (30)$$

$$\begin{aligned} E_y^n(i, j, k) &= g_{a,y}(i, j, k) \left(D_y^n(i, j, k) - I_y^{n-1}(i, j, k) \right), \\ I_y^n(i, j, k) &= I_y^{n-1}(i, j, k) + g_{b,y}(i, j, k) E_y^n(i, j, k), \end{aligned} \quad (31)$$

$$\begin{aligned} E_z^n(i, j, k) &= g_{a,z}(i, j, k) \left(D_z^n(i, j, k) - I_z^{n-1}(i, j, k) \right), \\ I_z^n(i, j, k) &= I_z^{n-1}(i, j, k) + g_{b,z}(i, j, k) E_z^n(i, j, k). \end{aligned} \quad (32)$$

Rule: per-component indexed form of the ADE-based flux update; matches the discrete relations used in the implementation for E_x, E_y, E_z and I_x, I_y, I_z .

1.5 Magnetic field: from Faraday's law to explicit update

1. Continuous equation. Faraday's law (μ constant):

$$\frac{\partial H_x}{\partial t} = -\frac{1}{\mu} \left(\frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} \right). \quad (33)$$

Rule: $(\nabla \times \mathbf{E})_x = \partial E_z / \partial y - \partial E_y / \partial z$.

2. Substitute. Central difference in time and space (Yee; H_x at half-integer positions in y, z). Implementation order: H is updated after E in the same step, so the curl uses $\mathbf{E}^{n+1}$:

$$\frac{H_x^{n+1/2} - H_x^{n-1/2}}{\Delta t} = -\frac{1}{\mu} \left(\frac{E_z^{n+1}(i, j+1, k+\frac{1}{2}) - E_z^{n+1}(i, j, k+\frac{1}{2})}{\Delta y} - \frac{E_y^{n+1}(i, j+\frac{1}{2}, k+1) - E_y^{n+1}(i, j+\frac{1}{2}, k)}{\Delta z} \right) \quad (34)$$

$$(35)$$

Rule: finite-difference substitution.

3. Explicit update for H_x .

$$H_x^{n+\frac{1}{2}}(i, j+\frac{1}{2}, k+\frac{1}{2}) = H_x^{n-\frac{1}{2}}(i, j+\frac{1}{2}, k+\frac{1}{2}) - \frac{\Delta t}{\mu} \left(\frac{E_z^{n+1}(i, j+1, k+\frac{1}{2}) - E_z^{n+1}(i, j, k+\frac{1}{2})}{\Delta y} - \frac{E_y^{n+1}(i, j+\frac{1}{2}, k+1) - E_y^{n+1}(i, j+\frac{1}{2}, k)}{\Delta z} \right). \quad (36)$$

Rule: solve for $H_x^{n+1/2}$; uses $\mathbf{E}^{n+1}$ (H updated after E in the same step). Analogous expressions hold for H_y and H_z using the other two components of $\nabla \times \mathbf{E}$. Perfectly Matched Layers (PML) are applied at boundaries by extending these updates with auxiliary variables and graded coefficients; the interior scheme is as above.

Explicit Yee-staggered form for all H components. Collecting the three components with full index staggering on the Yee grid (as in the thesis), the magnetic field updates can be written as

$$H_x^{n+\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}) = H_x^{n-\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2}) - \frac{\Delta t}{\mu} \left(\frac{E_z^{n+1}(i, j + 1, k + \frac{1}{2}) - E_z^{n+1}(i, j, k + \frac{1}{2})}{\Delta y} - \frac{E_y^{n+1}(i, j + \frac{1}{2}, k + 1) - E_y^{n+1}(i, j + \frac{1}{2}, k)}{\Delta z} \right), \quad (37)$$

$$H_y^{n+\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}) = H_y^{n-\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2}) - \frac{\Delta t}{\mu} \left(\frac{E_x^{n+1}(i + \frac{1}{2}, j, k + 1) - E_x^{n+1}(i + \frac{1}{2}, j, k)}{\Delta z} - \frac{E_z^{n+1}(i + 1, j, k + \frac{1}{2}) - E_z^{n+1}(i, j, k + \frac{1}{2})}{\Delta x} \right), \quad (38)$$

$$H_z^{n+\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k) = H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k) - \frac{\Delta t}{\mu} \left(\frac{E_y^{n+1}(i + 1, j + \frac{1}{2}, k) - E_y^{n+1}(i, j + \frac{1}{2}, k)}{\Delta x} - \frac{E_x^{n+1}(i + \frac{1}{2}, j + 1, k) - E_x^{n+1}(i + \frac{1}{2}, j, k)}{\Delta y} \right). \quad (39)$$

These are the fully indexed Yee-staggered forms corresponding to the generic Faraday-law discretization and are algebraically equivalent to (36) (and its y, z analogues).

1.6 SAR and heat source

1. Continuous. SAR (power per unit mass):

$$\text{SAR} = \frac{\sigma |\mathbf{E}|^2}{2\rho}, \quad |\mathbf{E}|^2 = E_x^2 + E_y^2 + E_z^2. \quad (40)$$

Rule: definition from time-averaged power deposition.

2. Discrete. Over N time steps, $|\mathbf{E}_{\text{rms}}|^2 = \frac{1}{N} (\sum_n E_x^2 + \sum_n E_y^2 + \sum_n E_z^2)$; then

$$\text{SAR}(i, j, k) = \frac{\sigma(i, j, k) |\mathbf{E}_{\text{rms}}|^2(i, j, k)}{2\rho(i, j, k)}. \quad (41)$$

Rule: same formula with RMS field. Volumetric heat source: $Q = \text{SAR} \cdot \rho$ (W/m³), so $Q(i, j, k) = \text{SAR}(i, j, k) \rho(i, j, k)$. Rule: one-way coupling from EM to thermal (thesis).

1.7 Thermal: Pennes to Gauss-Seidel

1. Continuous. Steady-state Pennes (no perfusion):

$$\nabla \cdot (k \nabla T) + Q = 0. \quad (42)$$

Rule: conservation of energy with SAR-derived source.

2. Discrete Laplacian. Piecewise-constant k per voxel; central difference for $\nabla^2 T$:

$$\frac{1}{(\Delta x)^2} (T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} + T_{i,j,k+1} + T_{i,j,k-1} - 6T_{i,j,k}) = -\frac{Q_{i,j,k}}{k_{i,j,k}}. \quad (43)$$

Rule: 7-point stencil, $k\nabla^2 T + Q = 0$.

3. Solve for $T_{i,j,k}$.

$$T_{i,j,k} = \frac{1}{6} \left(T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} + T_{i,j,k+1} + T_{i,j,k-1} + \frac{Q_{i,j,k} (\Delta x)^2}{k_{i,j,k}} \right). \quad (44)$$

Rule: Gauss-Seidel iteration; Dirichlet $T = T_{\text{boundary}}$ on the domain boundary (thesis, Thermal Solver Implementation).

1.8 Derivation chain

Continuous Ampere $\rightarrow$ (7) $\rightarrow$ (8) $\rightarrow$ (9) $\rightarrow$ (10).

Flux formulation: (23) $\rightarrow D$ updates (27)–(29) and E, I component-wise.

Continuous Faraday $\rightarrow$ (35) $\rightarrow$ (36) (and H_y, H_z).

SAR: (40) $\rightarrow$ (41); $Q = \text{SAR} \cdot \rho$.

Pennes (42) $\rightarrow$ (43) $\rightarrow$ (44).

2 Time Step (Courant Condition)

The time step is set from the Courant stability condition for a uniform grid ($\Delta x = \Delta y = \Delta z = dx$). Implemented at engine setup (e.g. `fdtd_brain_simulation_engine.py` around lines 2614–2621, and in `_run_minimal_fdtd_benchmark` 185–188).

$$\Delta t_{\text{courant}} = \frac{1}{c_0 \sqrt{1/(\Delta x)^2 + 1/(\Delta y)^2 + 1/(\Delta z)^2}}. \quad (45)$$

with $c_0 = 2.99792458 \times 10^8$ m/s (speed of light in vacuum).

$$\Delta t = F \cdot \Delta t_{\text{courant}}. \quad (46)$$

where F is the Courant factor (default 0.99), set by `-courant-factor`.

3 Material Coefficients (ADE for Lossy Media)

Lossy, dispersive-free materials use an auxiliary differential equation (ADE) in the engine. Relative permittivity ϵ_r , conductivity σ , and mass density ρ are assigned per tissue from a lookup table. The effective coefficients used in the E-field update are computed once per voxel (engine lines 2683–2691):

$$c = \frac{1}{\epsilon_r + \sigma \Delta t / \epsilon_0}, \quad \text{cond} = \frac{\sigma \Delta t}{\epsilon_0}. \quad (47)$$

with $\epsilon_0 = 8.854 \times 10^{-12}$ F/m. The arrays `eps_x`, `eps_y`, `eps_z` store c ; `conductivity_x/y/z` store cond .

4 FDTD Field Updates (Yee Scheme + PML)

4.1 D-field update (curl of $\mathbf{H}$)

Displacement field $\mathbf{D}$ is updated from $\nabla \times \mathbf{H}$ on a staggered grid. PML is applied via coefficients g_{i1}, g_{i2}, g_{i3} (and similarly g_{j*}, g_{k*}). Implemented in `calculate_dx_field`, `calculate_dy_field`, `calculate_dz_field` (lines 384–462).

Example for D_x (one component): with

$$(\nabla \times \mathbf{H})_x = \frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z},$$

the discrete curl at (i, j, k) is

$$\text{curl_h} = H_z|_{i,j,k} - H_z|_{i,j-1,k} - H_y|_{i,j,k} + H_y|_{i,j,k-1}. \quad (48)$$

The update uses PML auxiliary variable iD_x and coefficients $g_{j3}, g_{k3}, g_{j2}, g_{k2}, g_{i1}$:

$$iD_x \leftarrow iD_x + \text{curl_h}; \quad D_x \leftarrow g_{j3}g_{k3}D_x + g_{j2}g_{k2} \left(\frac{1}{2} \text{curl_h} + g_{i1} iD_x \right). \quad (49)$$

D_y and D_z follow from the other components of $\nabla \times \mathbf{H}$ with the corresponding PML indices.

4.2 E-field from D-field (ADE)

Electric field $\mathbf{E}$ is obtained from $\mathbf{D}$ and the integrated conduction current $\mathbf{I}$ (engine: `calculate_e_fields`, lines 484–514):

$$E_\alpha = c_\alpha (D_\alpha - I_\alpha), \quad I_\alpha \leftarrow I_\alpha + \text{cond}_\alpha E_\alpha. \quad (50)$$

for $\alpha \in \{x, y, z\}$, with c and cond from (47).

4.3 H-field update (curl of $\mathbf{E}$)

Magnetic field $\mathbf{H}$ is updated from $\nabla \times \mathbf{E}$ with PML (e.g. `calculate_hx_field`, `calculate_hy_field`, `calculate_hz_field`, lines 544–619). Example for H_x :

$$\text{curl_e} = E_y|_{i,j,k+1} - E_y|_{i,j,k} - E_z|_{i,j+1,k} + E_z|_{i,j,k}. \quad (51)$$

$$iH_x \leftarrow iH_x + \text{curl_e}; \quad H_x \leftarrow f_{j3}f_{k3}H_x + f_{j2}f_{k2} \frac{1}{2} (\text{curl_e} + f_{i1} iH_x). \quad (52)$$

f_* are the PML coefficients for the H-update (lines 299–357: cubic grading $x_n = 0.33 ((n_{\text{PML}} - n)/n_{\text{PML}})^3$).

5 SAR (Specific Absorption Rate)

5.1 Instantaneous SAR

Used per time step when accumulating E-field squared (`sar_instant`, `compute_sar_instantaneous`; see lines 762–784):

$$\text{SAR} = \frac{\sigma |\mathbf{E}|^2}{2\rho}, \quad |\mathbf{E}|^2 = E_x^2 + E_y^2 + E_z^2. \quad (53)$$

Units: W/kg. σ is averaged over x, y, z components when they differ (isotropic assumption).

5.2 SAR from time-averaged (RMS) E-field

After accumulating $\sum E_x^2$, $\sum E_y^2$, $\sum E_z^2$ over N_{samples} steps (`compute_sar`, lines 803–826):

$$|\mathbf{E}_{\text{rms}}|^2 = \frac{1}{N_{\text{samples}}} \left(\sum E_x^2 + \sum E_y^2 + \sum E_z^2 \right);$$

$$\text{SAR} = \frac{\sigma |\mathbf{E}_{\text{rms}}|^2}{2\rho}. \quad (54)$$

5.3 SAR from complex phasor (antenna optimization)

In the optimization path, SAR is computed from the synthesized complex field $\mathbf{E}_{\text{total}}$ (`compute_sar_from_complex_t`, lines 1077–1085):

$$\text{SAR} = \frac{\sigma |\mathbf{E}_{\text{total}}|^2}{2\rho}, \quad |\mathbf{E}_{\text{total}}|^2 = |\text{Re}(\mathbf{E}) + i \text{Im}(\mathbf{E})|^2. \quad (55)$$

6 Thermal Solver (Pennes Steady-State)

6.1 Heat source from SAR

Volumetric heat source (W/m^3) from SAR deposition (engine: $Q_{\text{heat}} = \text{SAR} \cdot \rho$, line 4358; see also comment at 4349):

$$Q = \text{SAR} \cdot \rho. \quad (56)$$

6.2 Steady-state bioheat (no perfusion)

Simplified Pennes equation with constant thermal conductivity k and no perfusion (`solve_steady_bioheat_3d`, docstring and implementation around 831–865):

$$\nabla \cdot (k \nabla T) + Q = 0. \quad (57)$$

Dirichlet boundary condition: $T = T_{\text{boundary}}$ (default 37°C) on the domain boundary.

6.3 Gauss-Seidel update

Central-difference Laplacian on a uniform grid with spacing dx ; Gauss-Seidel iteration over interior voxels with $k > 0$ (lines 844–861):

$$T_{i,j,k} = \frac{1}{6} \left(T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} + T_{i,j,k+1} + T_{i,j,k-1} + \frac{Q_{i,j,k} (\Delta x)^2}{k_{i,j,k}} \right). \quad (58)$$

Iteration stops when $\max |T - T_{\text{old}}| < \text{tol}$ (default 10^{-6}) or after a maximum number of iterations.

7 Antenna Optimization (J-ratio)

7.1 J-ratio objective

Definition of the optimization figure of merit (`compute_j_ratio`, lines 1089–1098):

$$J = \frac{\overline{\text{SAR}}_{\text{tumor}}}{\overline{\text{SAR}}_{\text{healthy}}}. \quad (59)$$

Overlines denote spatial mean over the respective tissue masks.

7.2 Penalized objective (optional)

When a penalty weight w is used to limit healthy-tissue hotspots (`compute_robust_objective`, 1102–1122):

$$J_{\text{eff}} = J - w \frac{P_{95}(\text{SAR}_{\text{healthy}})}{\overline{\text{SAR}}_{\text{tumor}}}. \quad (60)$$

P_{95} denotes the 95th percentile of SAR in the healthy-tissue region.

8 Source and Ramp

8.1 CW soft source with Gaussian ramp

For continuous-wave or sinusoidal sources, the soft source adds to D_z at the source gap (e.g. lines 1005–1008, 3622–3625). Ramp and value:

$$\text{ramp}(t) = 1 - \exp\left(-\frac{1}{2}\left(\frac{t}{t_{\text{ramp}}}\right)^2\right); \quad \text{source_val} = \text{ramp}(t) \cdot \sin(2\pi f_0 t \Delta t + \phi). \quad (61)$$

Default $t_{\text{ramp}} = 30$ time steps; ϕ is the phase offset (e.g. for 4-quadrant optimization).

9 Summary Table

Label	Quantity	Code location (approx.)
(45)	Courant time step	2618–2620, 185–188
(46)	Actual Δt	2621, 188
(47)	Effective ε , cond	2684–2690
(48), (49)	D-field update	384–462
(50)	E from D (ADE)	508–513
(51), (52)	H-field update	544–619
(53)	Instantaneous SAR	762–784
(54)	RMS SAR	803–826
(55)	SAR from phasor	1077–1085
(56)	Heat source Q	4358
(57)	Pennes steady-state	834–836
(58)	Gauss-Seidel T update	853–861
(59)	J-ratio	1089–1098
(60)	Penalized objective	1102–1122
(61)	CW ramp and source	1005–1008, 3622–3625